Certified discovery: statistically valid materials screening claims from a small DFT verification budget

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Abstract

Machine-learning models now screen millions of candidate crystals, and headline counts of newly discovered stable materials steer synthesis effort and public perception of artificial intelligence in science. Such counts are predictions, not measurements: they inherit the unknown bias of the model that produced them and carry no statistical guarantee. Here we show that discovery claims can be certified by pairing abundant model predictions with a small, randomly sampled budget of density functional theory verifications, yielding confidence intervals for the precision of a flagged candidate set, its count of truly stable materials, and yield differences between models, together with a closed-form rule for the required budget. Emulating verification campaigns for fifteen public energy models on roughly 257,000 exhaustively labeled candidates, naive counts overstate true discoveries by 10% to 124% at the exact-hull threshold; the overstatement shrinks with model accuracy but does not vanish, so even the single most accurate 2026 model still announces one phantom discovery in ten, while the older graph-network predictors announce more false discoveries than true ones. Five hundred verifications certify discovery rates to within 2.6–4.5 percentage points at nominal 95% coverage.

Introduction

High-throughput screening with machine-learned surrogates has become the dominant mode of computational materials discovery. Universal machine-learning interatomic potentials trained on public density functional theory (DFT) corpora[1,2], from early graph networks such as CHGNet[3], M3GNet[4] and MEGNet[40] to the current generation trained on OMat24 and OMol25[8,9], including MACE[5,6], MatterSim[7] and equivariant

transformer potentials such as eSEN[10], rank the thermodynamic stability of millions of hypothetical crystals at negligible cost, and generative pipelines built on them have announced discovery counts in the hundreds of thousands[12]. These totals carry weight well beyond the papers that report them, since they direct synthesis effort, shape the curation of community databases, and inform how funders and the public assess artificial intelligence in the physical sciences. Present reporting practice admits two modes. Small campaigns verify every selected candidate with DFT, which is exact but restricts studies to a few hundred candidates, whereas large campaigns rely on the surrogate alone, although a critical examination has found machine-learned formation energies to be substantially less reliable for stability classification than their regression errors suggest[13]. No intermediate practice has been established.

A count produced by a model, however, is only as trustworthy as the model itself. Independent assessments have questioned the novelty, stability and utility of large announced sets[14,15]; a re-screening this year of 894 previously reported stable materials retained only 298 as thermodynamically stable once competing phases were fully treated[16]; and the evaluation community has documented systematic optimism in surrogate stability calls, including softened energy landscapes[17] and inflated accuracy under test-set redundancy[18]. The community's response so far has been better benchmarks[18–20] and better uncertainty estimates for individual predictions[21,22]. Most recently, verification budgets and reliability certificates have entered the conversation: risk-model-guided DFT auditing has been proposed to recover discoveries a single surrogate would miss[23]. Yet none of these closes the gap to the quantity a discovery paper actually asserts. Benchmark scores, including the precision reported on evaluation corpora, describe a model on a reference distribution where a full label census happens to exist; they say nothing about the campaign at hand, which by definition screens an unlabeled pool, and even a benchmark precision of 0.9 implies tens of thousands of false discoveries at the scale of current announcements. Per-structure uncertainties do not aggregate into a valid confidence statement about a count, and risk-guided verification improves yield without making any claim statistically valid. What is missing is a guarantee about the aggregate content of a specific screened candidate set.

Recent statistical work supplies the missing bridge. Prediction-powered inference[24] constructs provably valid confidence intervals for population quantities by combining a small gold-standard sample with abundant machine predictions, correcting the prediction-based estimate with an empirically measured bias term, the rectifier. Refinements tune the reliance on predictions so that the interval is never asymptotically wider than the classical label-only interval[25], stratify the correction[26], share strength across many related estimation tasks[27], and connect the framework to conformal calibration[28]. Adjacent lines extend the idea to adaptive

label collection[29,30] and to cost-optimal allocation between raters of different price and quality[31], and a complementary family, conformal selection, controls the false discovery rate of a shortlist at selection time given a labeled calibration set exchangeable with the candidates[32,33]. These tools have been deployed in proteomics, surveys, drug-compound screening and language-model evaluation[24,26,32]; to our knowledge, none has been applied to computational materials screening, where the gold standard is expensive but perfectly automatable and campaigns publish precisely the aggregate claims these methods can certify.

Here we show that discovery claims can be certified: reported as confidence intervals anchored to a documented verification budget rather than as raw model counts. We adapt prediction-powered inference to the screening setting, where the estimand lives on a model-selected finite pool, and show that certification there acquires exact finite-population guarantees by identification with classical design-based estimation, whose equivalence with prediction-powered estimators has recently been articulated in general form[34]: unbiasedness and a closed-form variance at any fixed coefficient, an estimator variance smaller than the classical one's by exactly the factor $1-\rho^2$ at the optimal coefficient, where ρ is the pool-level prediction-label correlation, and a closed-form rule for the DFT budget a target precision requires. We prove that adaptively targeted verification is uncertifiable, formalizing why risk-ranked audits cannot back aggregate claims, and we introduce a paired certified-yield estimator that lets a single DFT budget compare many models on a common pool and provably dominates splitting that budget whenever the models' error patterns are nonnegatively correlated. Emulating certification campaigns on the WBM candidate set[35], exhaustively labeled and curated by Matbench Discovery[18], we quantify how far naive counts overstate certified discoveries, how many DFT verifications a certified claim requires, and whether model rankings survive certification. This contribution differs from each neighboring line of work in what it guarantees and for whom. Benchmarks score models on a reference corpus[18,20], and per-structure uncertainty methods attach error bars to individual predictions[21]; certification instead makes a statement about the aggregate content of one campaign's own pool. Conformal selection[32] decides what to shortlist before any verification and requires calibration data exchangeable with the candidates, whereas certification operates after selection, drawing its labels from the flagged pool itself and returning intervals for counts and model differences rather than a selection rule; the two occupy complementary stages of a single pipeline. Risk-guided verification[23] targets its DFT audits at the candidates judged most informative, which raises yield but forfeits valid aggregate inference because the audited sample is selected rather than representative; the present work identifies which verification designs preserve validity and certifies the claims those designs support. All estimators, the emulation harness and every figure are

reproducible with an open toolkit that runs on commodity hardware, so that any screening campaign, including generative pipelines, can report certified counts at no cost beyond the verifications themselves.

Results

Discovery counts are predictions, not measurements

A screening campaign applies a model f to a candidate universe and reports the flagged set $S = \{i : f \text{ calls } i \text{ stable}\}$ of size N (Fig. 1a). Let $y_i \in \{0,1\}$ indicate true stability under the campaign's own computational oracle, defined throughout as DFT energy above the convex hull $E_{\text{hull}} \leq \tau$ at a stated threshold τ; certification is relative to this oracle, not to experimental synthesizability, a scope we return to in the Discussion. The quantities campaigns assert are the discovery precision $\pi = \frac{1}{N}\sum_{i\in S} y_i$, the discovery count $N\pi$, and, when models are compared, the certified yield $q_f = \frac{1}{|U|}\sum_{i\in U} y_i f_i$ over a common pool U. The naive report, which counts every flagged candidate or averages predicted stabilities over the pool, is the imputation estimator of prediction-powered inference[24]: it admits no valid confidence interval, and its error equals the model's false-discovery behavior on the very set the model selected, which is exactly where surrogate models are least trustworthy[17]. Benchmark precision cannot substitute for certification, because leaderboard precision is computed against a complete label census on a fixed reference corpus, whereas a campaign, by definition, screens a pool for which no census exists, and a model's error profile need not transfer between the two[18,23].

A certification protocol from prediction-powered inference

Our protocol (Fig. 1c) draws a uniform random subsample $\mathscr{L} \subset S$ of size n, computes DFT labels on it, and forms the power-tuned prediction-powered estimate

$$\hat{\pi}_\lambda \;=\; \frac{\lambda}{N}\sum_{i\in S} f_i \;+\; \frac{1}{n}\sum_{i\in\mathscr{L}} \big(y_i - \lambda f_i\big), \tag{1}$$

whose first term trusts the predictions and whose second term, the rectifier, measures and removes their bias on the verified subsample. Setting $\lambda = 0$ recovers the classical label-only interval exactly, so certification strictly generalizes current practice; at the variance-optimal coefficient, and with variances known, the certified interval is narrower than the classical one by exactly $\sqrt{1-\rho^2}$ at every budget and confidence level, where

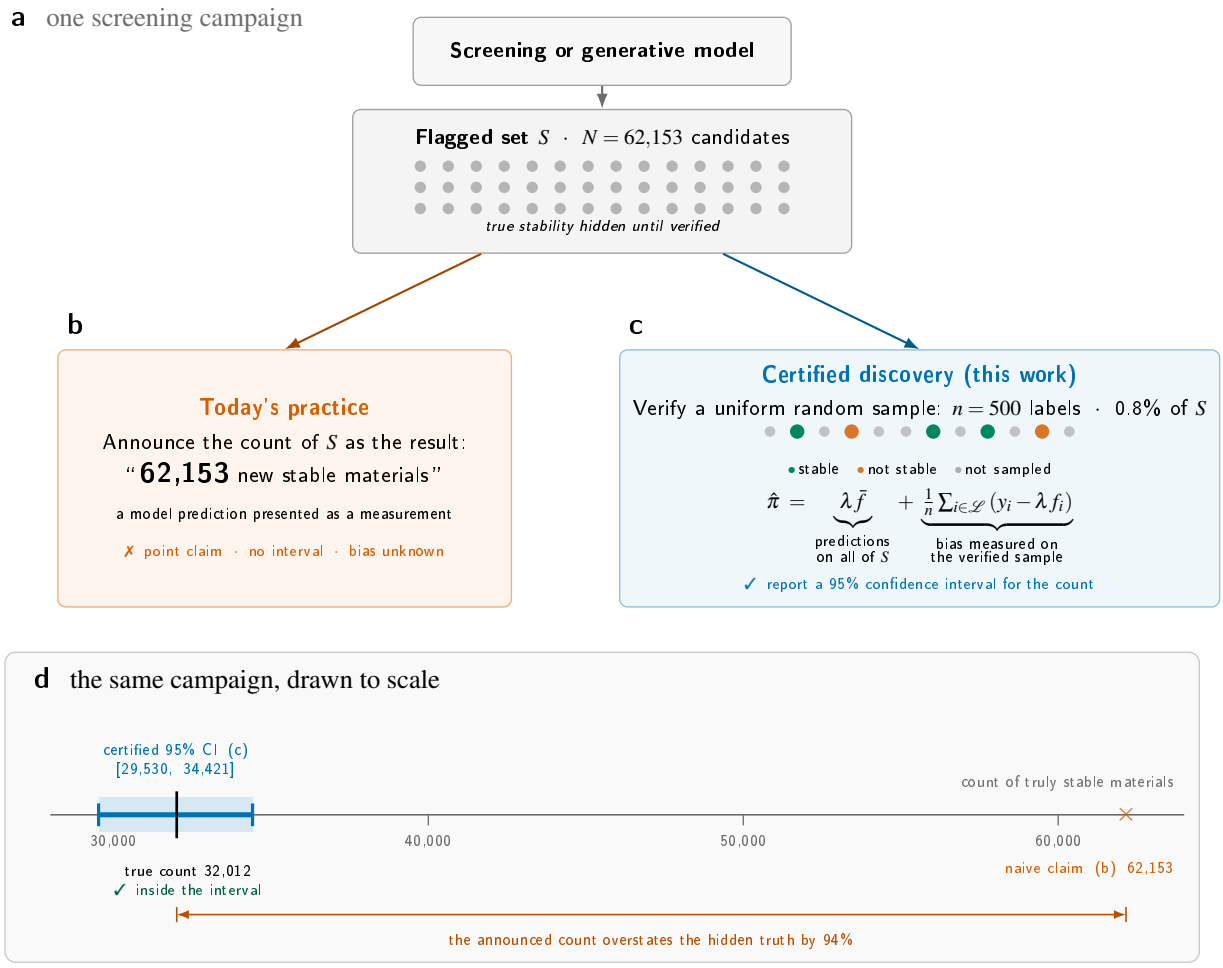


Fig. 1 Certified discovery replaces announced counts with confidence intervals backed by a small verification budget. Every number shown is one real campaign from Table 1 (CHGNet, exact-hull threshold). **a**, A screening or generative model flags a candidate set S of $N = 62{,}153$ candidates; the campaign cannot observe which flagged candidates are truly stable (gray). **b**, Today's practice announces the size of S as the discovery count: a prediction presented as a measurement, with no interval and a bias no one has measured. **c**, Certified discovery reveals DFT labels for a uniform random sample of $n = 500$ flagged candidates, 0.8% of S, then combines the model's predictions on all of S with the bias measured on the verified sample (equation (1)) and reports a confidence interval. **d**, The same campaign drawn to scale: the certified 95% interval [29,530, 34,421] contains the census truth of 32,012, hidden from both branches, while the naive claim of 62,153 overstates it by 94%.

ρ is the pool-level correlation between predictions and labels; with the coefficient and variances estimated from the subsample, the interval is never asymptotically wider (Methods, Theorem 1 and Proposition 1). Model selection of the pool does not bias the procedure, because the estimand is defined on the selected pool and the verified subsample is drawn from that same pool; what is not permitted is recycling verifications acquired elsewhere; the accompanying toolkit's interface admits only pool-sampled labels and documents this requirement at every entry point, though no software can verify the provenance of the labels it is given. Counts follow by scaling; a closed-form rule converts a target precision into the required DFT budget (Methods,

Corollary 1); and a paired construction compares the certified yields of many models on a common pool with a single verification budget, provably beating an equal split of that budget whenever the models' error patterns are nonnegatively correlated (Methods, Proposition 3). The added cost of certification over current practice is the DFT budget prescribed by Corollary 1 plus one random-number seed; how small that budget is in practice is quantified per model in Fig. 3a,b.

The protocol is valid and efficient on a synthetic positive control

Before treating any real campaign, we validate the estimators in a setting where ground truth is known by construction: a synthetic pool of $N = 20{,}000$ candidates flagged by an optimistically biased screener. The naive count overstates precision by $+0.20$ absolute (0.573 reported against a true 0.375), while classical and prediction-powered intervals both achieve their nominal 95% coverage (0.955 and 0.952 over 400 replicates) and the prediction-powered interval is uniformly narrower at every budget (Fig. 2c); all four numbers are printed by the released notebook that also generates the figure. These numbers are properties of the synthetic construction and serve as a positive control for the software; the real campaigns follow below. The efficiency law survives contact with reality: overlaying the fifteen real WBM campaigns on the same axes, each model's measured pool correlation against its realized certified-to-classical width ratio at $n = 400$ falls on the $\sqrt{1-\rho^2}$ curve to within 0.002 (Fig. 2d), even though the deployed estimator plugs in estimated rather than known variances (Proposition 1).

Certified discovery counts for public energy models

Using the exhaustive DFT labels of the WBM corpus as a hidden oracle, we emulate certification campaigns for fifteen models spanning the Matbench Discovery leaderboard, from the current accuracy frontier down to the earliest graph-network energy predictors: the model's flagged-stable pool at $\tau \in \{0, 0.05, 0.10\}$ eV/atom defines S, labels are revealed only for a random subsample of size $n = 500$, and the certified count is compared with the naive count and the exhaustive truth. Table 1 reports the results. The census columns are now final: naive counts overstate true discoveries by 10.2% (PET-OAM-XL) to 124.5% (CGCNN) at $\tau = 0$ on the full pool, and the overstatement rises with formation-energy error overall (compare the MAE and overstatement columns of Table 1). The gradient is steep and the message survives the passage to current models: the ten frontier potentials, whose formation-energy errors cluster near 0.02 eV/atom, overstate by 10–21% (mean

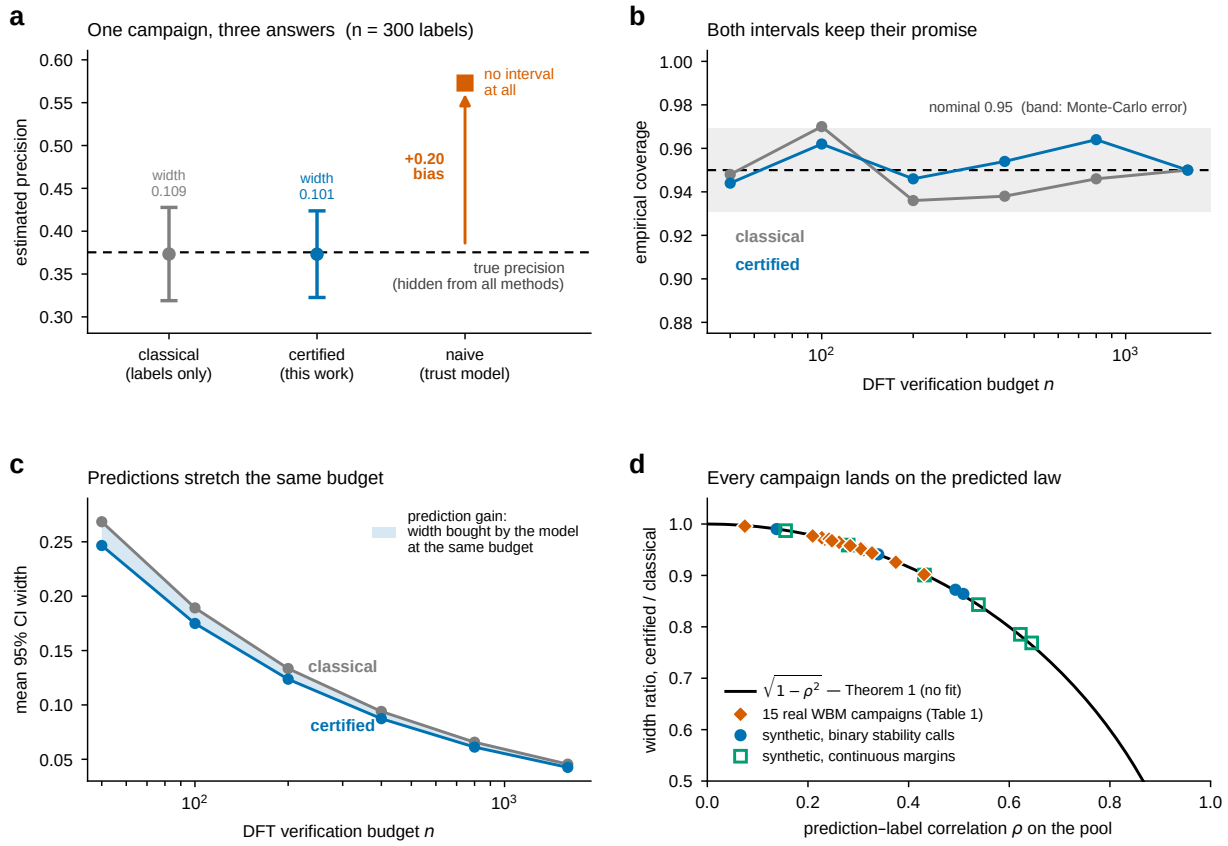


Fig. 2 The certification estimators are calibrated, efficient, and follow the efficiency law predicted by the theory, on a synthetic pool with ground truth known by construction. All panels use a pool of $N = 20{,}000$ candidates flagged by an optimistically biased screener (Methods). **a**, At a budget of $n = 300$, the classical and certified intervals both cover the true precision (dashed line), the certified interval is tighter, and the naive model-trusting estimate overstates precision by $+0.20$ with no interval at all. **b**, Empirical coverage of nominal 95% intervals across budgets (500 Monte Carlo replicates per point); the shaded band is the Monte Carlo error of the coverage estimate itself, and both methods stay within it. **c**, Mean interval width against budget; the shaded wedge is the width the predictions buy at fixed cost. **d**, Empirical certified-to-classical width ratio against the pool-level prediction–label correlation ρ, for binary stability calls (filled circles) and continuous stability margins (open squares), each swept over prediction-noise levels (150 replicates per point); the solid curve is the known-variance law $\sqrt{1-\rho^2}$ of Theorem 1. Binary calls cap the attainable correlation near $\rho \approx 0.5$ in this construction, whereas continuous margins reach further along the law, which is why the toolkit accepts margins. Diamonds are the fifteen real WBM campaigns of Table 1 ($\tau = 0$, full pool): each model's realized width ratio at $n = 400$ (mean over 500 replicates) against its measured pool correlation lies on the same curve, within 0.002 for every model.

14%), whereas the five older graph networks overstate by 94–124%. Naive optimism therefore shrinks as models improve but does not disappear at the frontier: even the single most accurate model in this set, EquiformerV3+DeNS-OAM at 0.017 eV/atom, announces 4,073 phantom discoveries, an overstatement of 10.3%, while the weakest announce more false discoveries than true ones. The ordering is not strictly monotonic in regression error, because classification calibration at the stability threshold matters beyond mean accuracy, as the closely spaced frontier rows show. Two robustness checks are equally final. On

Table 1 Certified versus naive discovery claims for fifteen public machine-learning energy models on the WBM pool. Threshold $\tau = 0$ eV/atom; rows ordered by naive overstatement. Flagged, true, overstatement and precision-gap entries are exact census quantities, computed against the full WBM labels and independent of any sampling; they are final. Interval and coverage entries use a verification budget of $n = 500$ labels per campaign and 1,000 Monte Carlo replicates. Horizontal bars visualize the overstatement column (blue, current generation; vermillion, earlier graph networks); the two groups are separated by a roughly threefold gap in accuracy.

	Accuracy	Discovery count			Naive error		Validity
Model	MAE[a]	Flagged[b]	True[c]	Certified 95% CI[d]	Overstatement (%)[e]	Precision gap[f]	Coverage[g]
Current generation (OMat24/OMol25-era training corpora)							
PET-OAM-XL	0.018	42,619	38,664	[37,614, 39,714]	10.2	0.093	0.944
EquiformerV3+DeNS-OAM	0.017	43,437	39,364	[38,332, 40,479]	10.3	0.094	0.931
eSEN-30M-OAM	0.018	42,814	38,722	[37,689, 39,815]	10.6	0.096	0.946
MatRIS-10M-OAM	0.019	42,958	38,783	[37,708, 39,867]	10.8	0.097	0.955
ORB v3	0.023	43,497	38,305	[37,192, 39,545]	13.6	0.119	0.956
MatterSim v1 5M	0.024	39,624	34,700	[33,640, 35,826]	14.2	0.124	0.951
SevenNet-MF-ompa	0.021	44,484	38,556	[37,373, 39,883]	15.4	0.133	0.955
GRACE-2L-OAM	0.023	42,859	36,960	[35,771, 38,214]	16.0	0.138	0.953
DPA3-v1-OpenLAM	0.023	43,638	37,329	[36,057, 38,579]	16.9	0.145	0.942
MACE-MPA-0	0.028	43,477	36,055	[34,709, 37,407]	20.6	0.171	0.952
Earlier graph networks (2018–2023)							
CHGNet	0.061	62,153	32,012	[29,530, 34,421]	94.2	0.485	0.953
ALIGNN	0.092	56,632	28,060	[25,649, 30,494]	101.8	0.505	0.946
MEGNet	0.128	54,380	24,811	[22,427, 27,144]	119.2	0.544	0.963
M3GNet	0.072	75,829	33,967	[30,820, 36,925]	123.2	0.552	0.952
CGCNN	0.135	57,037	25,411	[23,095, 27,889]	124.5	0.554	0.934

[a]Formation-energy mean absolute error (eV/atom) of the published predictions against DFT on the same filtered corpus ($|\text{error}| \leq 5$ eV/atom); overstatement rises with error overall, though not strictly monotonically. [b]Size of the model's flagged-stable pool at τ; this is the count a naive campaign would announce. [c]Census count from the exhaustive WBM labels (hidden from the estimators; used only for evaluation). [d]Median-replicate interval for the count, $N\times$ equation (1) with PPI++ tuning. [e]$(\text{flagged} - \text{true})/\text{true} \times 100$. [f]Naive precision (1 by construction of the flagged pool) minus true precision. [g]Fraction of 1,000 replicates whose certified interval contains the true count; nominal 0.95, Monte Carlo s.e. 0.007.

the redundancy-controlled unique-prototype pool the overstatement persists at 7.8%–130.2%, so test-set leakage does not explain it, and loosening the threshold to $\tau = 0.05$ and 0.10 eV/atom shrinks the range to 5.2%–54.2% and 4.0%–31.0% respectively: naive optimism is concentrated exactly at the exact-hull threshold that discovery announcements use. The certified intervals deliver their guarantee: every interval in Table 1 contains the census count, and across all ninety (model, threshold, pool) configurations the empirical coverage of the nominal 95% intervals ranges from 0.92 to 0.97 with median 0.95 (Fig. 3c and Supplementary Table 1), within Monte Carlo precision of nominal at 1,000 replicates.

One verification budget certifies rankings across models

The paired estimator compares certified yields on the union of two models' flagged pools using a single label sample per replicate. Table 2 reports all fourteen pairs adjacent in the error-based (formation-energy MAE)

Table 2 Do model rankings survive certification? Certified-yield comparisons for adjacent leaderboard pairs under one shared verification budget. Paired estimator of Proposition 3 on the union pool: $n = 500$ shared labels per replicate, 200 replicates, medians reported; Holm-corrected across all pairs. No reversal of the error-based order is statistically supported: every interval that excludes zero matches that order.

	Certified yield difference, $A - B$				Error-based order	
Model pair, A vs. B	$\widehat{\Delta q}$[a]	95% CI	vs. zero[b]	Holm-adj. p[c]	ΔMAE[d]	Agreement[e]
Adjacent pairs within the current generation						
EquiformerV3+DeNS-OAM vs. eSEN-30M-OAM	$+0.013$	$[-0.000, +0.027]$		0.297	0.001	n.s.
eSEN-30M-OAM vs. PET-OAM-XL	$+0.002$	$[-0.013, +0.017]$		1.000	0.000	n.s.
PET-OAM-XL vs. MatRIS-10M-OAM	-0.002	$[-0.018, +0.014]$		1.000	0.000	n.s.
MatRIS-10M-OAM vs. SevenNet-MF-ompa	$+0.005$	$[-0.011, +0.022]$		1.000	0.002	n.s.
SevenNet-MF-ompa vs. DPA3-v1-OpenLAM	$+0.025$	$[+0.007, +0.043]$		0.062	0.002	agrees
DPA3-v1-OpenLAM vs. ORB v3	-0.020	$[-0.038, -0.001]$		0.262	0.000	reversed*
ORB v3 vs. GRACE-2L-OAM	$+0.027$	$[+0.009, +0.046]$		**0.047**	0.000	agrees
GRACE-2L-OAM vs. MatterSim v1 5M	$+0.046$	$[+0.024, +0.068]$		$<$ **0.001**	0.001	agrees
MatterSim v1 5M vs. MACE-MPA-0	-0.028	$[-0.051, -0.005]$		0.137	0.004	reversed*
Across the generation gap						
MACE-MPA-0 vs. CHGNet	$+0.054$	$[+0.028, +0.080]$		$<$ **0.001**	0.033	agrees
Adjacent pairs among the earlier graph networks						
CHGNet vs. M3GNet	-0.023	$[-0.045, +0.001]$		0.297	0.011	n.s.
M3GNet vs. ALIGNN	$+0.063$	$[+0.034, +0.091]$		$<$ **0.001**	0.020	agrees
ALIGNN vs. MEGNet	$+0.039$	$[+0.010, +0.069]$		0.079	0.037	agrees
MEGNet vs. CGCNN	-0.008	$[-0.037, +0.021]$		1.000	0.007	n.s.

[a]Difference in certified yield $q_A - q_B$ over the union of the two flagged pools (Methods). [b]The interval drawn against the zero line (gray): blue, Holm-significant; gray, not significant; vermillion, reversed sign that does not survive Holm correction. [c]Two-sided, Holm-corrected across all pairs in the table. [d]$\mathrm{MAE}_B - \mathrm{MAE}_A$ (eV/atom); every pair is ordered so A has the lower error, and larger error gaps are easier to separate. [e]"agrees": the certified interval excludes zero on the side matching the error-based order; "n.s.": the interval contains zero. "reversed*": the raw interval excludes zero on the side opposite the error-based order, but the pair does not survive Holm correction, so the reversal is not statistically supported. Bold p-values are significant at the 5% level after Holm correction.

order. Four pairs separate significantly after Holm correction, and every one of them matches the error-based order. The remaining ten adjacent pairs are statistically indistinguishable at a 500-label budget; two of them carry a reversed point-estimate sign whose raw interval excludes zero, but neither survives Holm correction, so no reversal of the error-based order is statistically supported. The reversals that fail to reach significance fall among the frontier models, whose formation-energy errors differ by a thousandth of an electronvolt per atom or less, precisely the regime the theory predicts a 500-label budget cannot resolve. The practical reading, pre-registered in our protocol, is symmetric: error-based rankings survive certification wherever the data can distinguish neighbors, and fine-grained leaderboard orderings among adjacent frontier models are not certified claims at realistic verification budgets.

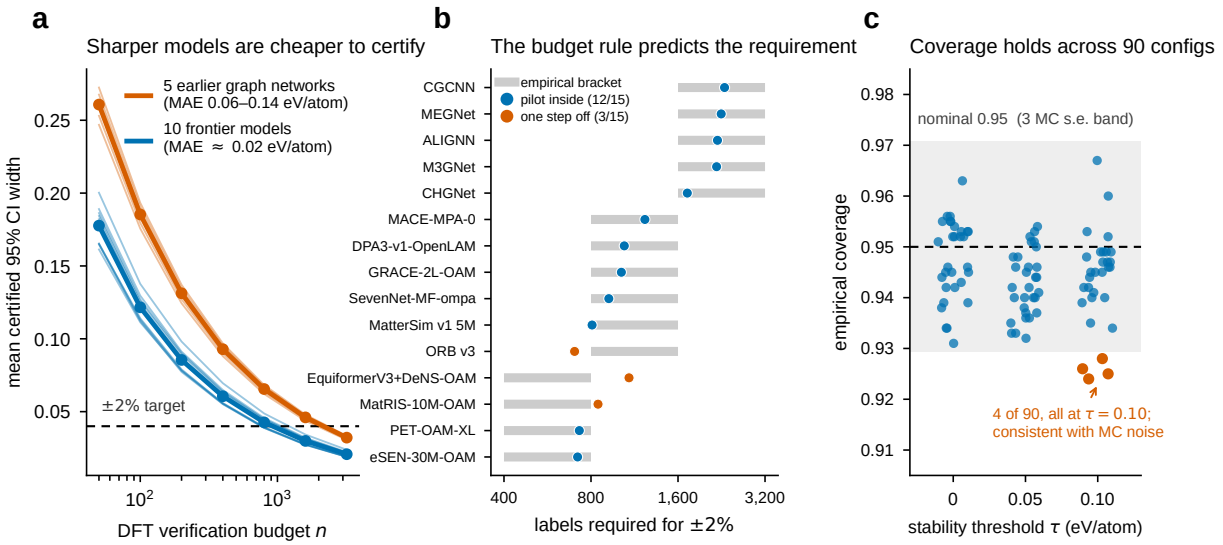


Fig. 3 Verification budgets and calibration on the WBM pool. **a**, Mean certified 95% interval width against verification budget at $\tau = 0$ (full pool, 500 replicates per point). Thin lines are individual models, thick lines group means: the ten frontier potentials (blue) and the five earlier graph networks (vermillion) separate cleanly, because a sharper model's flagged pool has higher precision and therefore lower base variance (Theorem 1(iii)), making the same certification cheaper; the dashed line marks the $\pm 2\%$ width target. **b**, The closed-form budget rule of Corollary 1, evaluated from a 100-label pilot, against the empirical requirement per model: gray bars bracket the true crossing between the last budget that misses the $\pm 2\%$ target and the first that meets it; the pilot prediction lands inside the bracket for twelve of the fifteen models (blue) and is one grid step off for the other three (vermillion): optimistic for ORB v3 and one step high for EquiformerV3+DeNS-OAM and MatRIS-10M-OAM, the scatter expected when a 100-label pilot estimates the rectifier variance. **c**, Empirical coverage of the nominal 95% certified intervals for all ninety (model, threshold, pool) configurations at $n = 500$ and 1,000 replicates, grouped by threshold. Eighty-six configurations lie within three Monte Carlo standard errors of nominal (shaded band); the four below it (vermillion) all occur at the loosest threshold $\tau = 0.10$, at rates consistent with Monte Carlo fluctuation, and every certified interval in Table 1 nonetheless contains its census count.

Optimal spending of the verification budget

Two practical questions govern adoption. First, how many DFT verifications certify a claim to a given precision? Figure 3a answers this per model: 500 labels certify precision to within 2.6–4.5 percentage points across the fifteen models (mean certified half-width; the median-replicate intervals of Table 1 are equal or tighter), and the $\pm 2\%$ target is reached by 800 labels for the sharpest frontier models, whose high-precision flagged pools have the smallest base variance, and by 1,600–3,200 for the rest, in every case under 6% of the pool a naive campaign would otherwise trust unverified. Corollary 1 answers the same question in closed form from a 100-label pilot estimate of the rectifier variance, and its prediction falls inside the empirical bracketing interval, between the last budget that misses the target and the first that meets it, for twelve of the fifteen models (Fig. 3b); for the remaining three the pilot-based prediction misses by one budget grid step in either

direction (optimistic for ORB v3, one step high for EquiformerV3+DeNS-OAM and MatRIS-10M-OAM), the scatter expected when the pilot variance estimate is noisy (Supplementary Note 2). Second, where should the verifications go? The stratified estimator with Neyman allocation (Methods) concentrates labels where the model's bias is most variable; stratifying by model-confidence tertile reduced interval width by at most 8% (GRACE-2L-OAM) and by under 1% for the other models tested (Supplementary Table 2), so we report stratification as an optional refinement rather than a source of large gains in this setting.

Discussion

Certified discovery reframes a cultural problem as a budgetary one. The field does not need to stop using surrogate models; it needs to attach a randomly assigned verification budget, sized by Corollary 1, to each campaign and report intervals instead of counts. Because the verified subsample doubles as an audit trail, the protocol also makes discovery claims contestable in a way raw counts are not: a skeptical reader can re-examine the sampled structures, re-run the oracle on them, and recompute the interval.

Three limitations bound our claims. First, certification is relative to the DFT settings that define the labels; published assessments place high-throughput DFT formation energies on the order of 0.1 eV/atom from experiment[2], so a certified-stable count is a statement about the computational hull, not about synthesizability[15]; Supplementary Note 4 states precisely which claims are and are not certified. The protocol is oracle-agnostic and applies unchanged with any automatable higher-fidelity oracle, and with experimental oracles up to their own measurement error, at whatever per-label cost those imply. Second, validity requires that the verified subsample be drawn randomly from the reported pool. Campaigns whose verifications were selected adaptively, for example by verifying the most promising or highest-risk candidates first[23], cannot be certified retroactively by this construction, because the audited sample is no longer representative of the pool. Extending certification to adaptive and sequential verification, where generation and verification interleave, is the natural next step; active statistical inference[29,30] and cost-optimal rater allocation[31] provide the validity-preserving tools, multi-task prediction-powered constructions[27] suggest how a single budget could share strength across models and strata, and pairing certification with a prospective conformal-selection stage[32] would give one pipeline both FDR control at selection time and certified counts at reporting time. Third, our empirical scope is inorganic crystal stability on one exhaustively labeled corpus; the estimators are domain-generic, but the empirical magnitudes we report are not automatically portable, and we deliberately

re-run all analyses on the redundancy-controlled unique-prototype pool[18] to guard against test-set leakage inflating the apparent value of predictions.

Within these bounds, the protocol converts announced discovery totals from unfalsifiable point estimates into auditable statistical statements at a cost any screening campaign can afford, and the same construction applies wherever an expensive automatable oracle meets abundant machine predictions, from catalyst screening to protein design.

Methods

Notation, estimands and assumptions

Let S be the flagged pool, $|S| = N$, with fixed but unknown labels $y_1, \ldots, y_N \in \{0,1\}$ and model outputs $f_1, \ldots, f_N \in [0,1]$, which may be binary stability calls or monotone-transformed stability margins; the estimator is valid for either and tighter for informative margins. The estimand is the finite-population mean $\pi = \frac{1}{N}\sum_{i\in S} y_i$, with counts $N\pi$ and paired yields as defined in Results. Write $\bar{f} = \frac{1}{N}\sum_{i\in S} f_i$ and, with $\frac{1}{N-1}$ normalization, the population variances S_y^2, S_f^2 and covariance S_{yf} over S; when $S_y, S_f > 0$, let $\rho = S_{yf}/(S_y S_f)$ denote the pool-level prediction-label correlation. Three assumptions govern validity. (A1) Sampling design: $\mathscr{L}$ is a uniform simple random sample without replacement of size $n \geq 2$ from S, or a stratified such sample, drawn independently of the model outputs conditional on S. (A2) Asymptotic regime: interval validity is asymptotic in n via the finite-population central limit theorem; for the small-n, rare-positive regime (fewer than ten labeled members of either class) the toolkit falls back to $\lambda = 0$ with a Wilson score interval, omitting the finite-population correction so the fallback is conservative (Supplementary Note 3), and coverage is checked empirically in all experiments. (A3) Oracle semantics: labels are the campaign's stated oracle, $E_{\text{hull}} \leq \tau$ under fixed DFT settings; all guarantees are relative to it.

Exact certification theory

The following results are stated for the precision estimand; counts follow by scaling every quantity by N; paired yields are treated in Proposition 3. All randomness is the sampling design of A1, conditional on the pool: labels and predictions are fixed numbers attached to the pool's members.

Theorem 1 (Exact finite-population certification)**.** *Fix $\lambda \in \mathbb{R}$ and let $\hat{\pi}_\lambda$ be the estimator of equation* (1).

The statements below are exact algebraic consequences of Assumption A1 for arbitrary fixed real attributes $\{(y_i, f_i)\}_{i\in S}$*, of which the campaign setting* $y_i \in \{0,1\}$*,* $f_i \in [0,1]$ *is the special case used in Results and the real-valued case is used by Proposition 3. For* $2 \le n \le N$*: (i)* $\mathbb{E}[\hat{\pi}_\lambda] = \pi$*; (ii)* $\mathrm{Var}(\hat{\pi}_\lambda) = \frac{1}{n}\left(1 - \frac{n}{N}\right) S_r^2(\lambda)$ *with* $S_r^2(\lambda) = S_y^2 - 2\lambda S_{yf} + \lambda^2 S_f^2$*; (iii)* $\lambda = 0$ *recovers the classical label-only estimator, and if* $S_f^2 > 0$ *the unique variance-minimizing coefficient is* $\lambda^\star = S_{yf}/S_f^2$*, at which* $\mathrm{Var}(\hat{\pi}_{\lambda^\star}) = \frac{1}{n}\left(1 - \frac{n}{N}\right)\left(S_y^2 - S_{yf}^2/S_f^2\right)$*, equal to* $\frac{1}{n}(1 - \frac{n}{N})(1-\rho^2)S_y^2$ *whenever* $S_y > 0$*: with known variances, the certified interval is narrower than the classical one by the factor* $\sqrt{1-\rho^2}$ *exactly, at every budget n and every confidence level; (iv) if* $n = N$ *then* $\hat{\pi}_\lambda = \pi$ *for every* λ*.*

Proof. Write $r_i = y_i - \lambda f_i$ and $\bar{r} = \frac{1}{N}\sum_{i\in S} r_i = \pi - \lambda \bar{f}$. The prediction term $\lambda \bar{f}$ is deterministic given S, and the rectifier is the mean of a simple random sample without replacement from the finite population $\{r_i\}_{i\in S}$, hence unbiased for $\bar{r}$ with variance $\frac{1}{n}(1 - \frac{n}{N})S_r^2$, the standard without-replacement formula[36]; adding $\lambda \bar{f}$ gives (i) and (ii). Expanding $S_r^2(\lambda)$ bilinearly gives the stated quadratic, whose unique minimizer for $S_f^2 > 0$ is $\lambda^\star = S_{yf}/S_f^2$ with minimum $S_y^2 - S_{yf}^2/S_f^2 = (1-\rho^2)S_y^2$, and $\lambda = 0$ gives $S_r^2(0) = S_y^2$, proving (iii). For (iv), at $n = N$ the rectifier equals $\bar{r}$ exactly, so $\hat{\pi}_\lambda = \lambda \bar{f} + \pi - \lambda \bar{f} = \pi$. □

Lemma 1 (Exact interval construction)**.** *Fix* λ *and let* $s_r^2(\lambda)$ *be the sample variance, with* $n-1$ *divisor, of the labeled residuals* $\{y_i - \lambda f_i\}_{i\in\mathscr{L}}$*; because the predictions are observed on all of S, the quantities* $\bar{f}$ *and* S_f^2 *are known exactly and require no estimation. Define*

$$\hat{v}(\lambda) \;=\; \frac{1}{n}\Big(1 - \frac{n}{N}\Big)\, s_r^2(\lambda) \;+\; \frac{\lambda^2 S_f^2}{N}.$$

Under A1 with $2 \le n \le N$*,* $\mathbb{E}[s_r^2(\lambda)] = S_r^2(\lambda)$*, hence* $\mathbb{E}[\hat{v}(\lambda)] = \mathrm{Var}(\hat{\pi}_\lambda) + \lambda^2 S_f^2/N \;\ge\; \mathrm{Var}(\hat{\pi}_\lambda)$ *at every n, with relative excess* $O(n/N)$*; consequently the reported interval* $\hat{\pi}_\lambda \pm z_{1-\alpha/2}\sqrt{\hat{v}(\lambda)}$ *is built from an exactly upward-biased variance estimate, its asymptotic coverage under A2 is at least* $1-\alpha$*, and in the certification regime* $n \ll N$ *the width inflation is second order. The role of the additive term when* λ *is estimated is quantified in Proposition 1.*

Proof. Under simple random sampling without replacement the sample variance of any fixed attribute is unbiased for the population variance with $N-1$ divisor[36]; applying this to the fixed residual population $\{y_i - \lambda f_i\}_{i\in S}$ gives $\mathbb{E}[s_r^2(\lambda)] = S_r^2(\lambda)$, and combining with Theorem 1(ii) gives the expectation identity. The inequality holds because $\lambda^2 S_f^2/N \ge 0$, and the ratio of the additive term to $\mathrm{Var}(\hat{\pi}_\lambda)$ equals $\frac{n}{N-n}\cdot\frac{\lambda^2 S_f^2}{S_r^2(\lambda)} =$

$O(n/N)$ whenever $S_r^2(\lambda) > 0$. □

Remark 1 (Lineage and what is new)**.** *At $\lambda = 1$ the estimator is the classical survey difference estimator, and at $\lambda^\star$ the design-based regression estimator[36,37]; the general equivalence between prediction-powered estimators and this survey lineage has recently been set out by Mozer[34]. The contribution of Theorem 1 is therefore not the estimator but the identification: because a screening campaign's flagged pool is itself the population its headline claim refers to, certification lives in the design-based finite-population regime, where the guarantees of (i)–(iv) are exact at every n; infinite-population prediction-powered inference[24,25] targets a distributional estimand from two samples, so even at a fixed coefficient its variance carries an additional term from the unlabeled sample, its tuned never-wider guarantee is asymptotic, and its estimand is not the realized content of any particular pool. The identification also grounds validity in nine decades of survey-sampling theory rather than in newly minted assumptions.*

Corollary 1 (Certification budget in closed form)**.** *Fix a working coefficient λ, a confidence level $1-\alpha$ with normal quantile $z = z_{1-\alpha/2}$, and a target half-width $w > 0$. The minimal budget with $z\sqrt{\mathrm{Var}(\hat{\pi}_\lambda)} \le w$ is*

$$n^\star(w) = \left\lceil \frac{N z^2 S_r^2(\lambda)}{N w^2 + z^2 S_r^2(\lambda)} \right\rceil,$$

clipped to $[2, N]$. As $N \to \infty$ this recovers the classical $z^2 S_r^2 / w^2$; at $\lambda^\star$, $S_r^2 = (1-\rho^2) S_y^2$, so better models need smaller budgets, with the reduction approaching the factor $1-\rho^2$ as $N \to \infty$ and being strictly more modest at finite N.

Proof. By Theorem 1(ii), $z^2 \mathrm{Var}(\hat{\pi}_\lambda) = z^2 S_r^2 (\frac{1}{n} - \frac{1}{N})$ is strictly decreasing in n on $[2, N]$ and hits w^2 at $n = N z^2 S_r^2 / (N w^2 + z^2 S_r^2)$; the least integer no smaller than this value is $n^\star(w)$. The limit and the $\lambda^\star$ substitution are immediate. □

In practice S_r^2 is unknown; the released toolkit estimates it from a small uniform pilot, and the budget-sweep experiment of Fig. 3b quantifies the accuracy of the resulting budgets against empirical width curves.

Proposition 1 (Plug-in coefficient)**.** *Let $\hat{\lambda}$ be the sample regression slope on $\mathscr{L}$,*

$$\hat{\lambda} = \widehat{S}_{yf} / \widehat{S}_f^2, \tag{2}$$

where $\widehat{S}_{yf}$ and $\widehat{S}_f^2$ are the sample covariance and variance on $\mathscr{L}$ with $n-1$ divisor, and $\hat{\lambda} := 0$ if $\widehat{S}_f^2 = 0$.

Under A1–A3 with $n \to \infty$, $N-n \to \infty$, $\liminf S_y^2 > 0$, $\liminf S_f^2 > 0$ and $\limsup \rho^2 < 1$, the estimator $\hat{\pi}_{\hat{\lambda}}$ has the same limiting distribution as $\hat{\pi}_{\lambda^\star}$; the studentized interval $\hat{\pi}_{\hat{\lambda}} \pm z_{1-\alpha/2}\sqrt{\hat{v}(\hat{\lambda})}$, with $\hat{v}$ the estimator of Lemma 1 whose deliberately conservative additive term also absorbs the variability of $\hat{\lambda}$, attains coverage at least $1-\alpha$ asymptotically; and by Theorem 1(iii) its asymptotic width never exceeds the classical interval's. This is the classical asymptotic theory of the design-based regression estimator under simple random sampling without replacement, established via the Hájek finite-population central limit theorem[36,37]; Supplementary Note 1 records the argument specialized to our bounded setting, whose key step is the exact identity $\hat{\pi}_{\hat{\lambda}} - \hat{\pi}_{\lambda^\star} = (\hat{\lambda} - \lambda^\star)(\bar{f} - \bar{f}_{\mathscr{L}}) = O_p(n^{-1})$.

Proposition 2 (Adaptive verification is uncertifiable)**.** *Let $z = z_{0.975}$. For every even $N \geq 8$ and every budget $4 \leq n \leq N/2$ there exist a pool $\{(y_i, f_i)\}_{i \in S}$ with true precision $\pi = \frac{1}{2}$ and a verification ordering, measurable with respect to side information available to the verifier, such that the classical, Wilson and prediction-powered 95% intervals computed as if A1 held are all contained in $(\frac{1}{2}, 1]$ and therefore have coverage zero. In particular, no procedure that computes intervals by the A1 formulas retains any nominal coverage level uniformly over pools once labels are revealed in an informative deterministic order.*

Proof. Take $y_i = 1$ for $i \leq N/2$ and $y_i = 0$ otherwise, $f_i \equiv 1$ (the whole pool is flagged), and any ordering that ranks the stable half first; orderings positively associated with correctness are precisely what confidence- and risk-ranked verification produce by design. The first $n \leq N/2$ revealed labels all equal 1. The classical estimator is 1 with zero sample variance, giving the degenerate interval $\{1\}$. The Wilson interval at $\hat{p} = 1$ is $[n/(n+z^2), 1]$, and $n/(n+z^2) > \frac{1}{2}$ whenever $n > z^2 \approx 3.84$, i.e. for all $n \geq 4$. The prediction-powered interval reduces to the classical one because $\widehat{S}_f^2 = 0$ forces $\hat{\lambda} = 0$. Each interval excludes $\pi = \frac{1}{2}$, so coverage is 0. □

Remark 2. *The construction is not exotic: any audit whose order is informative about correctness biases the revealed rectifier toward zero, which is the explicit objective of risk-ranked verification[23]. Proposition 2 is the formal reason certification requires the randomized design of A1, and why adaptively audited campaigns cannot be certified retroactively; validity-preserving adaptivity exists but must be built in from the start[29,30].*

Proposition 3 (Pairing dominates budget splitting)**.** *Let U be a common pool of size M, let models A, B have fixed outputs f^A, f^B on U, fix a working coefficient λ, and write the per-item yield rectifiers $r_i^A = y_i f_i^A - \lambda f_i^A$ and $r_i^B = y_i f_i^B - \lambda f_i^B$, with population variances S_A^2, S_B^2 and covariance S_{AB} over U. Consider two designs*

with the same total budget n (even, $n \le M$): the paired design draws one simple random sample of size n from U and estimates the yield difference with the estimator of equation (1)*, taking $y_i(f_i^A - f_i^B)$ as the outcome and $f_i^A - f_i^B$ as the prediction, so that its rectifier terms are $d_i = (y_i - \lambda)(f_i^A - f_i^B)$; the split design draws two independent simple random samples of size $n/2$ and estimates each yield separately at the same λ. Both are unbiased for $q_A - q_B$, and*

$$\mathrm{Var}_{\mathrm{paired}} = \tfrac{1}{n}\big(1 - \tfrac{n}{M}\big)\big(S_A^2 + S_B^2 - 2S_{AB}\big), \qquad \mathrm{Var}_{\mathrm{split}} = \tfrac{2}{n}\big(1 - \tfrac{n}{2M}\big)\big(S_A^2 + S_B^2\big).$$

If $S_{AB} \ge 0$ then $\mathrm{Var}_{\mathrm{paired}} < \mathrm{Var}_{\mathrm{split}}$ whenever $S_A^2 + S_B^2 > 0$.

Proof. Unbiasedness and both variance formulas follow from Theorem 1(i)–(ii) with $S = U$ and $N = M$, applied with outcome $y_i(f_i^A - f_i^B)$ and prediction $f_i^A - f_i^B$ for the paired design, whose rectifier population is $\{d_i\}$ with $d_i = r_i^A - r_i^B$, and applied to each yield separately for the split design, where independence of the two samples gives additivity. If $S_{AB} \ge 0$ then $S_A^2 + S_B^2 - 2S_{AB} \le S_A^2 + S_B^2$, and $1 - \frac{n}{M} < 2 - \frac{n}{M} = 2(1 - \frac{n}{2M})$ always; multiplying the two bounds gives strict inequality when $S_A^2 + S_B^2 > 0$. □

Models trained on overlapping corpora err on similar chemistries, making $S_{AB} \ge 0$ the expected regime; when a pilot indicates $S_{AB} < 0$, the two variance formulas decide the design directly. In the released implementation λ is estimated per design as in Proposition 1, and multiplicity across model pairs is Holm-corrected.

Stratified certification and budget allocation

Partition S into strata of sizes N_k and apply equation (1) within each; the stratified estimator is the N_k/N-weighted combination, with variance the weighted sum of Theorem 1(ii) applied per stratum. For a total budget n, the variance-minimizing allocation is Neyman's classical rule $n_k \propto N_k S_{r,k}$ applied to the per-stratum rectifier standard deviations[36,37], estimated from a small uniform pilot with a floor of two labels per stratum; the derivation and the two-stage pilot variant are standard and recorded in Supplementary Note 2. The materials-specific content is the choice of strata (chemical systems, structural prototypes, or model-confidence bins) and the rectifier-targeted pilot, evaluated empirically in Supplementary Table 2.

Data

All experiments use public artifacts only: the WBM candidate set with exhaustive DFT E_{hull} labels[35] in the curation of Matbench Discovery[18], and the per-model prediction files published on the Matbench Discovery leaderboard. The fifteen models evaluated span the current accuracy frontier and the earliest graph-network predictors. The ten current-generation models, most trained on OMat24 or OMol25[8,9], are EquiformerV3+DeNS-OAM[11], eSEN-30M-OAM[10], PET-OAM-XL, GRACE-2L-OAM, ORB v3, SevenNet-MF-ompa[38], MACE-MPA-0[5,6], MatRIS-10M-OAM, DPA3-v1-OpenLAM and MatterSim v1 5M[7]; the five earlier models are CHGNet[3], M3GNet[4], ALIGNN[41], MEGNet[40] and CGCNN[42]. Model display names, prototype identifiers and the exact prediction-file versions are recorded in the run metadata released with the toolkit. Because every pool member is labeled, verification campaigns of any budget can be emulated by revealing labels for random subsamples, and empirical coverage can be computed exactly against the census truth; no new DFT is required for this study.

Emulation protocol

For each model and threshold $\tau \in \{0, 0.05, 0.10\}$ eV/atom, and for both the full WBM pool and the redundancy-controlled unique-prototype pool[18], campaigns are emulated at label budgets $n \in \{50, 100, 200, 400, 800, 1600, 3200\}$, with replicate-indexed random seeds derived from a documented base seed. Monte Carlo replicate counts are 1,000 for the coverage analyses, 500 per point for the budget sweeps and 200 for the paired ranking comparisons, and are restated alongside each result. Reported quantities are empirical coverage against census truth, mean interval width, naive overstatement, and, for rankings, Holm-corrected pairwise significance. The synthetic positive control (a pool of 20,000 candidates with a latent-score screener and Bernoulli labels; construction fully specified in the released notebook) is run first, and the real-data experiments proceed only if it passes. Robustness variants perturb τ by ± 10 meV/atom and substitute continuous stability margins for binary calls, reported for a representative model in Supplementary Table 3.

Statistics and reproducibility

No new experimental data were generated; all randomness enters through documented seeds, and every figure and table is reproduced by a single notebook run from a clean checkout, a property verified end-to-end in an

independent environment against the archived source data. Confidence intervals are two-sided at the stated level; coverage claims are Monte Carlo estimates whose replicate counts are given alongside each result.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used are public: the WBM candidate set and DFT labels, and the per-model prediction files of the Matbench Discovery leaderboard. Exact versions, access commands and checksums are pinned in the released repository at https://github.com/hasankurban/certified-discovery. Source data for each main-text figure are provided in that repository under `results/`: Figure 1 from `table1_raw.csv` (the CHGNet, exact-hull row); Figure 2 from `fig2_synthetic_summary.csv`, `fig2_budget_curve_synthetic.csv` and `budgets_summary.csv`; and Figure 3 from `budget_curves.csv`, `budgets_summary.csv`, `model_mae.csv` and `table1_raw.csv`. Each file carries an SHA-256 checksum in the run manifest, and the figure-generation script `build_figures.py` reads exactly these files.

Code availability

A numpy-only toolkit implementing every estimator, the coverage harness, the analysis notebook, and the figure generators are available at https://github.com/hasankurban/certified-discovery and will be archived with a DOI on acceptance.

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Author contributions

Both authors contributed to all aspects of the work: conception, methodology, theory, software, experiments, analysis, and writing. Both authors reviewed and approved the manuscript and take responsibility for all content.

Competing interests

The authors declare no competing interests.